{i}DD

From words to wisdom

How LLMs and vector databases revolutionize data understanding



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22.10.2025

iteratec

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architect @ iteratec









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Truth is Subjective

Inquisition Maxim

LLMs hallucinate, how to minimize that effect

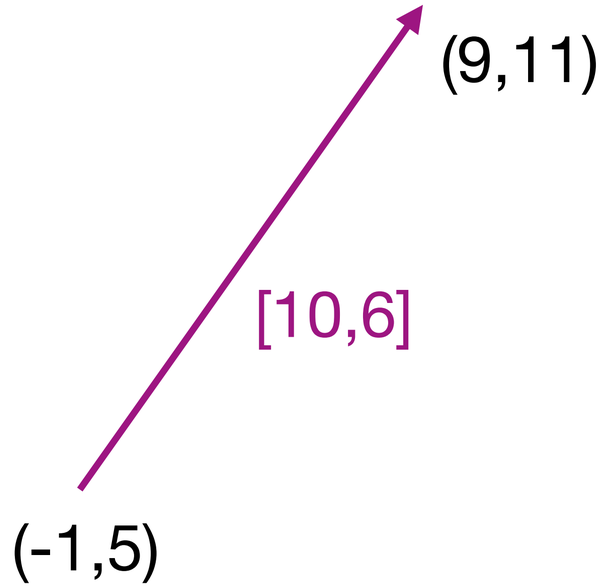
what are vectors
what is text embedding
what are vector databases
let's build a rag



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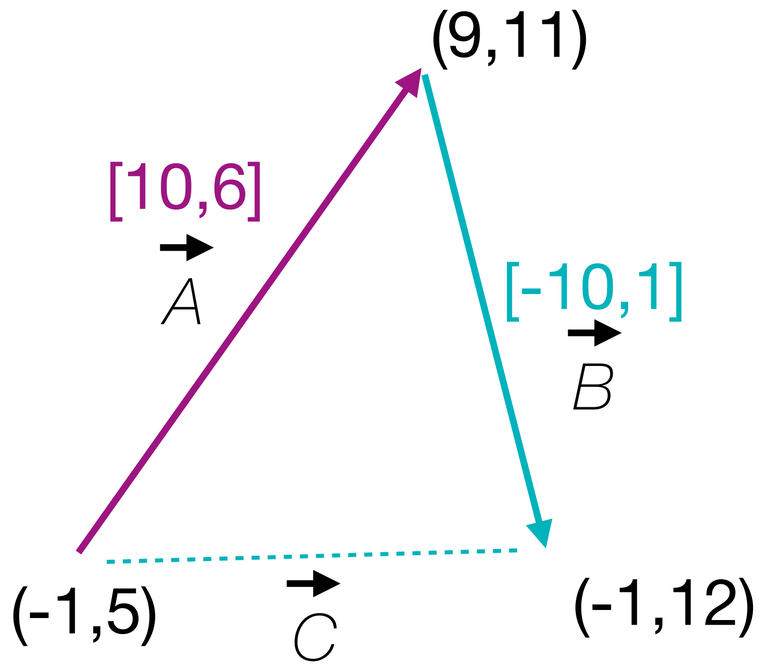
vector

properties



vector

arithmetics



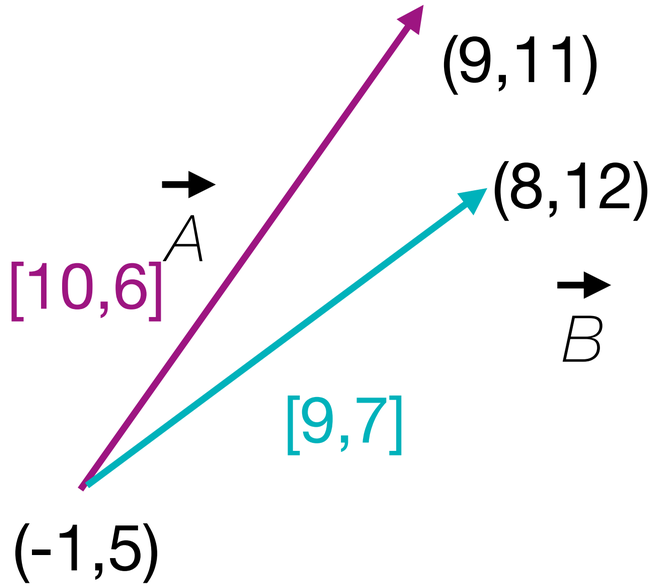
$$\vec{A} + \vec{B} = \vec{C} \quad [0,7]$$

$$\vec{C} - \vec{B} = \vec{A}$$

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vector

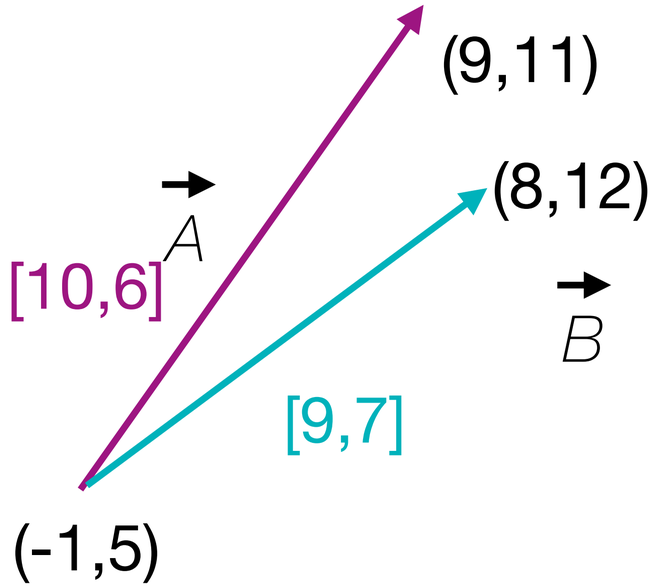
dot product



$$\vec{A} \times \vec{B} = a_1 b_1 + a_2 b_2 = 132$$

vector

cosine similarity

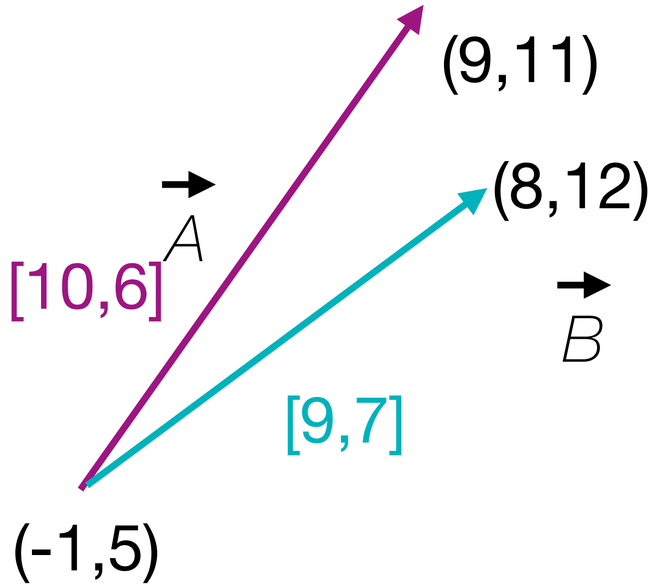


$$\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|}$$

132 / 11.66 × 11.4 = 0.99

vector

euclidean distance



$$d(A,B) = \sqrt{(A_1 - B_1)^2 + (A_2 - B_2)^2}$$

$$d(A,B) = \sqrt{(9-10)^2 + (7-6)^2} = \sqrt{1+1} = \sqrt{2} = 1{,}41$$

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vector

dimensions

[10, 21]

dimensions = 2

[5.59, 42, 3.21]

dimensions = 3

...

[42.231, 432.23, …, 20.213]

dimensions = n

embedding

tokens

The sky is blue

The sky is blueish

embedding

Embeddings are numerical representations, which make it easy for computers to understand the relationship.

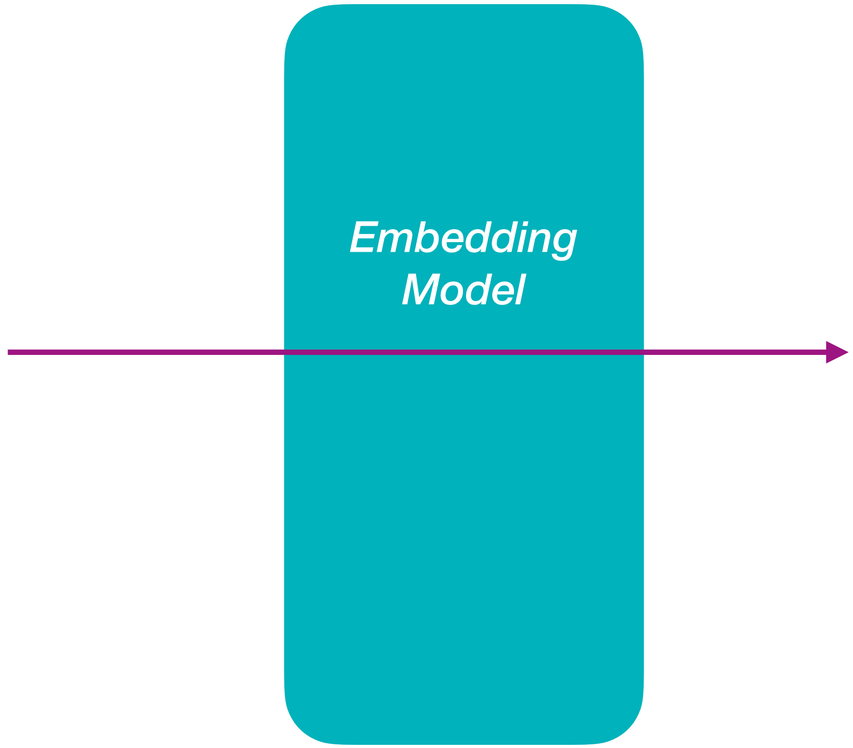
Embeddings that are numerically similar are also semantically similar.



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embedding

In the grim darkness of the far future, there is only war.




```
[
-0.0006528618978336453,
-0.03336622938513756,
-0.004293263889849186,
-0.017323806881904602,
-0.0009665339603088796,
0.013140465132892132
...
-0.021770961582660675,
0.006146245170384645 5,
-0.024811107665300 37,
-0.034019481390714645,
0.020276013761758804,
0.010891761630773544,
-0.009465908631682396,
0.006306418217718601,
0.005037596914917231
]
```

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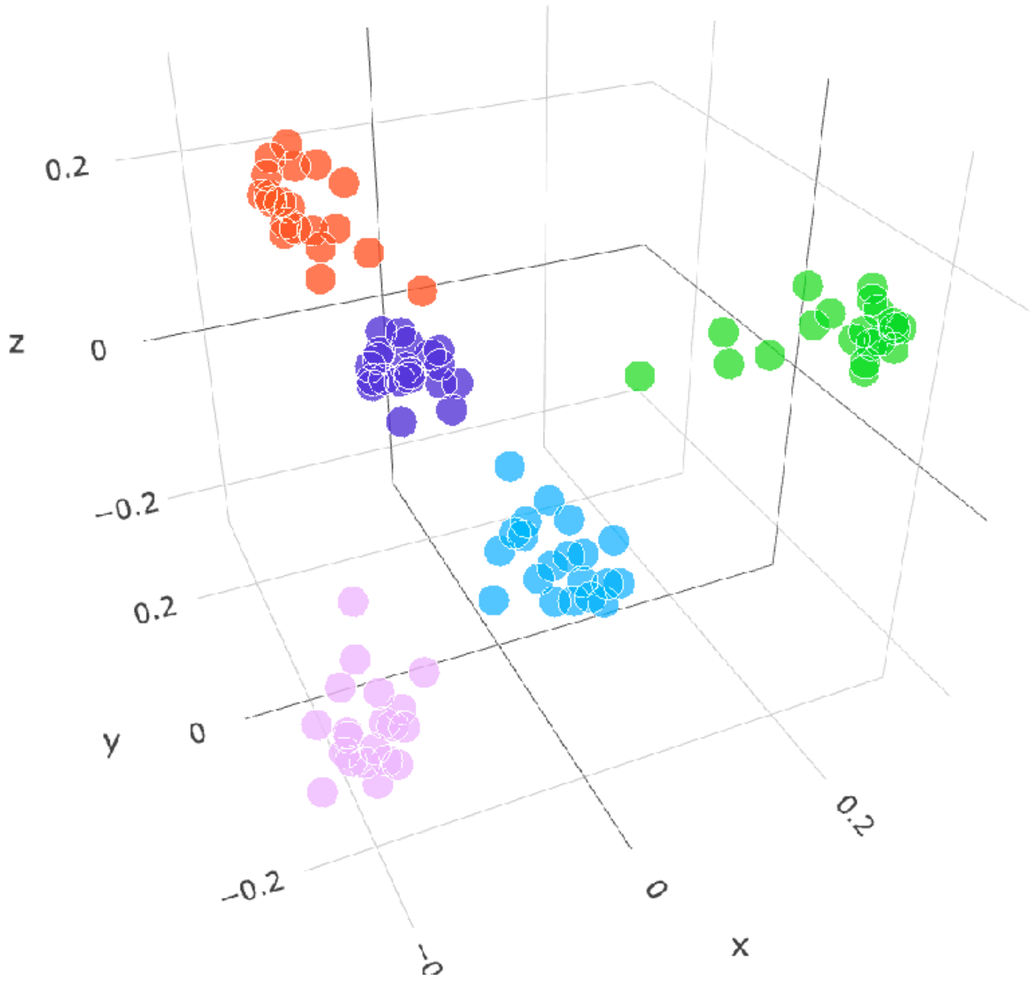
embedding

	living being	human	living at home	feline
Cat	0.92	0.13	0.65	0.8
Human	0.88	0.97	0.79	0.05
Tiger	0.9	0.13	-0.83	0.7



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embedding



animal

athlete

film

transportation

village



https://openai.com/index/introducing-text-and-code-embeddings/

Attention Is All You Need

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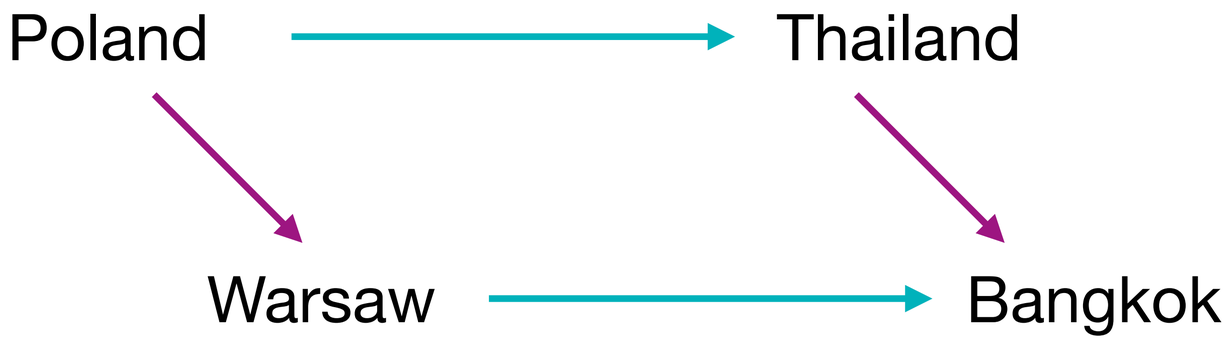
embedding

models

word2vec	*GloVe*	*BERT*	*...*
2013	*2014*	*2018*	

embedding

properties




```
v('Poland') - v('Warsaw') + v('Thailand') = v('Bangkok')
~ v('capital') + v('Thailand') = v('Bangkok')
```

embedding

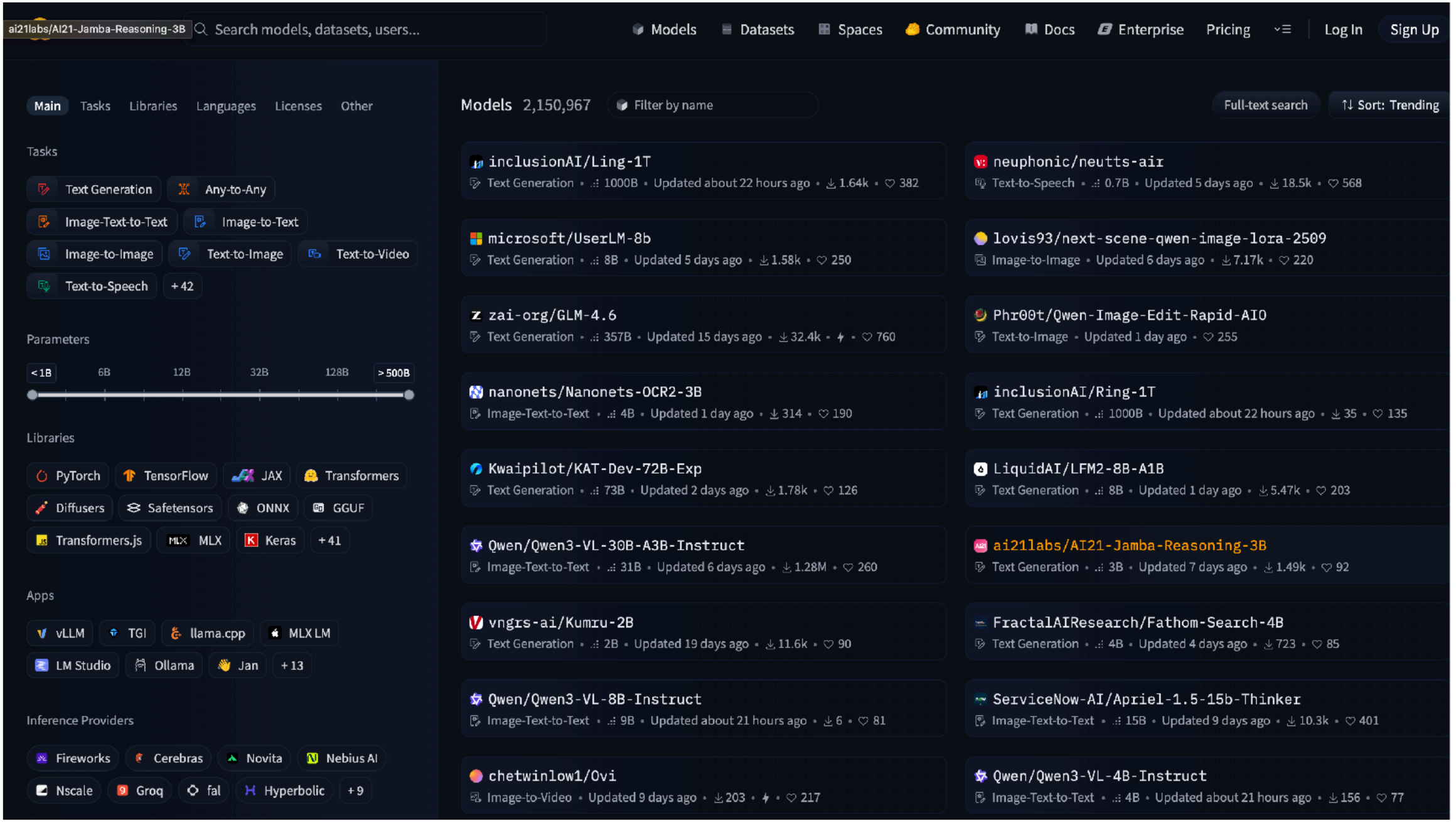
resources

Hugging Face

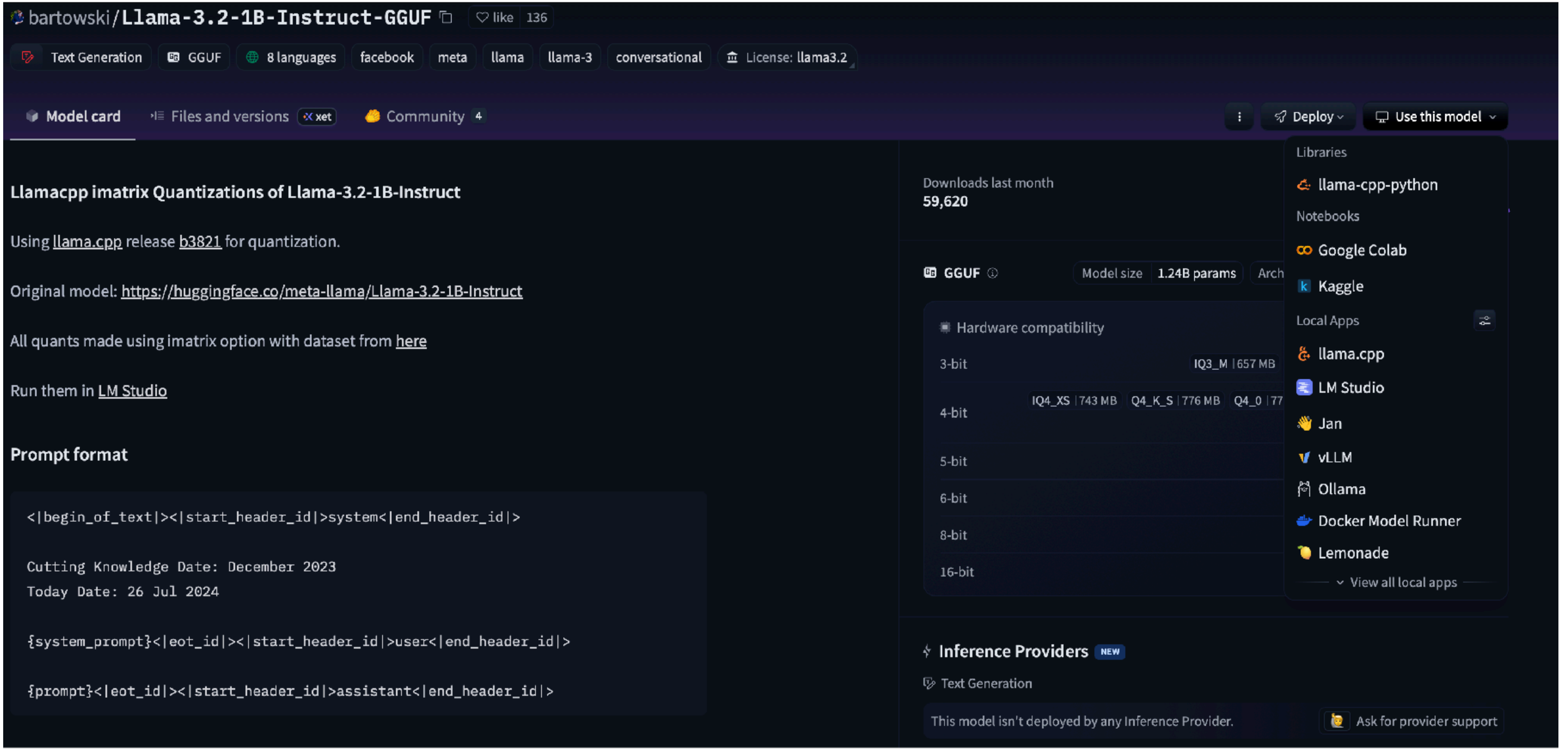
https://huggingface.co/

https://ollama.com/




ai21labs/AI21-Jamba-Reasoning-3B
Search models, datasets, users...
Models
Datasets
Spaces
Community
Docs
Enterprise
Pricing
Log In
Sign Up
Main
Tasks
Libraries
Languages
Licenses
Other
Tasks
Text Generation
Any-to-Any
Image-Text-to-Text
Image-to-Text
Image-to-Image
Text-to-Image
Text-to-Video
Text-to-Speech
+42
Parameters
<1B
6B
12B
32B
128B
>500B
Libraries
PyTorch
TensorFlow
JAX
Transformers
Diffusers
Safetensors
ONNX
GGUF
Transformers.js
MLX
Keras
+41
Apps
vLLM
TGI
llama.cpp
MLX LM
LM Studio
Ollama
Jan
+13
Inference Providers
Fireworks
Cerebras
Novita
Nebius AI
Nscale
Groq
fal
Hyperbolic
+9
Models 2,150,967
Filter by name
Full-text search
Sort: Trending
inclusionAI/Ling-1T
Text Generation • 1000B • Updated about 22 hours ago • 1.64k • 382
neuphonic/neutts-air
Text-to-Speech • 0.7B • Updated 5 days ago • 18.5k • 568
microsoft/UserLM-8b
Text Generation • 8B • Updated 5 days ago • 1.58k • 250
lovis93/next-scene-qwen-image-lora-2509
Image-to-Image • Updated 6 days ago • 7.17k • 220
zai-org/GLM-4.6
Text Generation • 357B • Updated 15 days ago • 32.4k • 760
Phr00t/Qwen-Image-Edit-Rapid-AIO
Text-to-Image • Updated 1 day ago • 255
nanonets/Nanonets-OCR2-3B
Image-Text-to-Text • 4B • Updated 1 day ago • 314 • 190
inclusionAI/Ring-1T
Text Generation • 1000B • Updated about 22 hours ago • 35 • 135
Kwaipilot/KAT-Dev-72B-Exp
Text Generation • 73B • Updated 2 days ago • 1.78k • 126
LiquidAI/LFM2-8B-A1B
Text Generation • 8B • Updated 1 day ago • 5.47k • 203
Qwen/Qwen3-VL-30B-A3B-Instruct
Image-Text-to-Text • 31B • Updated 6 days ago • 1.28M • 260
ai21labs/AI21-Jamba-Reasoning-3B
Text Generation • 3B • Updated 7 days ago • 1.49k • 92
vngrs-ai/Kumru-2B
Text Generation • 2B • Updated 19 days ago • 11.6k • 90
FractalAIResearch/Fathom-Search-4B
Text Generation • 4B • Updated 4 days ago • 723 • 85
Qwen/Qwen3-VL-8B-Instruct
Image-Text-to-Text • 9B • Updated about 21 hours ago • 6 • 81
ServiceNow-AI/Apriel-1.5-15b-Thinker
Image-Text-to-Text • 15B • Updated 9 days ago • 10.3k • 401
chetwinlow1/Ovi
Image-to-Video • Updated 9 days ago • 203 • 217
Qwen/Qwen3-VL-4B-Instruct
Image-Text-to-Text • 4B • Updated about 21 hours ago • 156 • 77

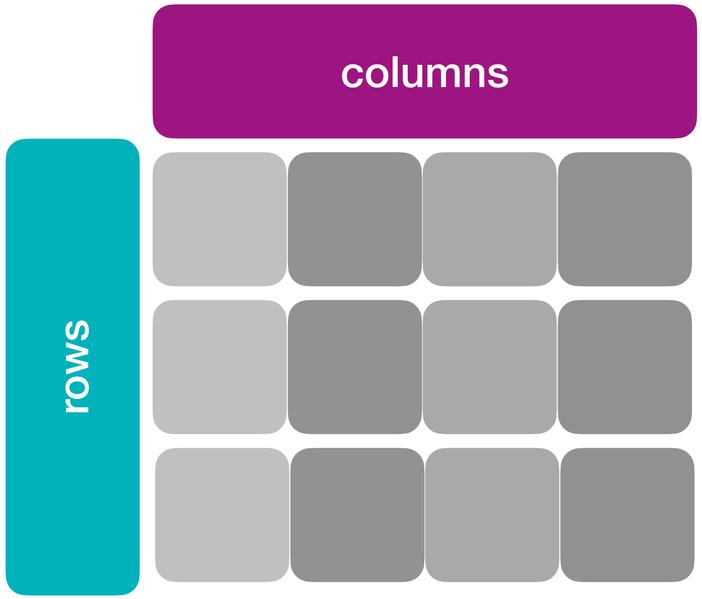
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bartowski/Llama-3.2-1B-Instruct-GGUF
like 136
Text Generation
GGUF
8 languages
facebook
meta
llama
llama-3
conversational
License: llama3.2
Model card
Files and versions
xet
Community 4
Deploy
Use this model
Llamacpp imatrix Quantizations of Llama-3.2-1B-Instruct
Using llama.cpp release b3821 for quantization.
Original model: https://huggingface.co/meta-llama/Llama-3.2-1B-Instruct
All quants made using imatrix option with dataset from here
Run them in LM Studio
Prompt format
<|begin_of_text|><|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
{system_prompt}<|eot_id|><|start_header_id|>user<|end_header_id|>
{prompt}<|eot_id|><|start_header_id|>assistant<|end_header_id|>
Downloads last month
59,620
GGUF
Model size
1.24B params
Hardware compatibility
3-bit
IQ3_M | 657 MB
4-bit
IQ4_XS | 743 MB
Q4_K_S | 776 MB
Q4_0 | 77
5-bit
6-bit
8-bit
16-bit
Libraries
llama-cpp-python
Notebooks
Google Colab
Kaggle
Local Apps
llama.cpp
LM Studio
Jan
vLLM
Ollama
Docker Model Runner
Lemonade
View all local apps
Inference Providers
NEW
Text Generation
This model isn't deployed by any Inference Provider.
Ask for provider support

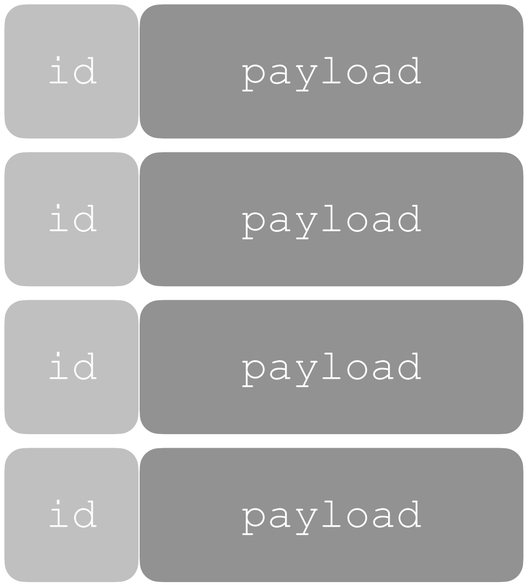
the database

Vector DB

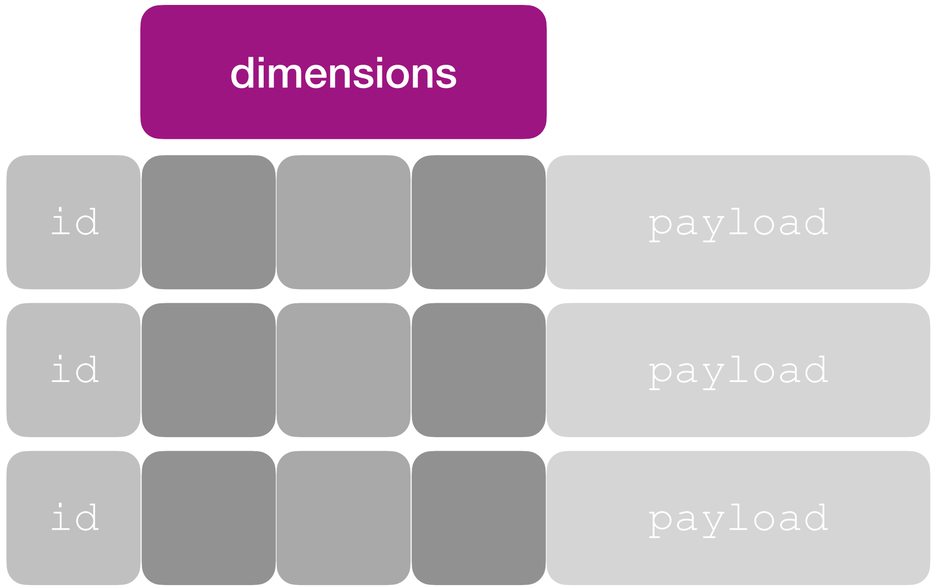
relation



document



vector



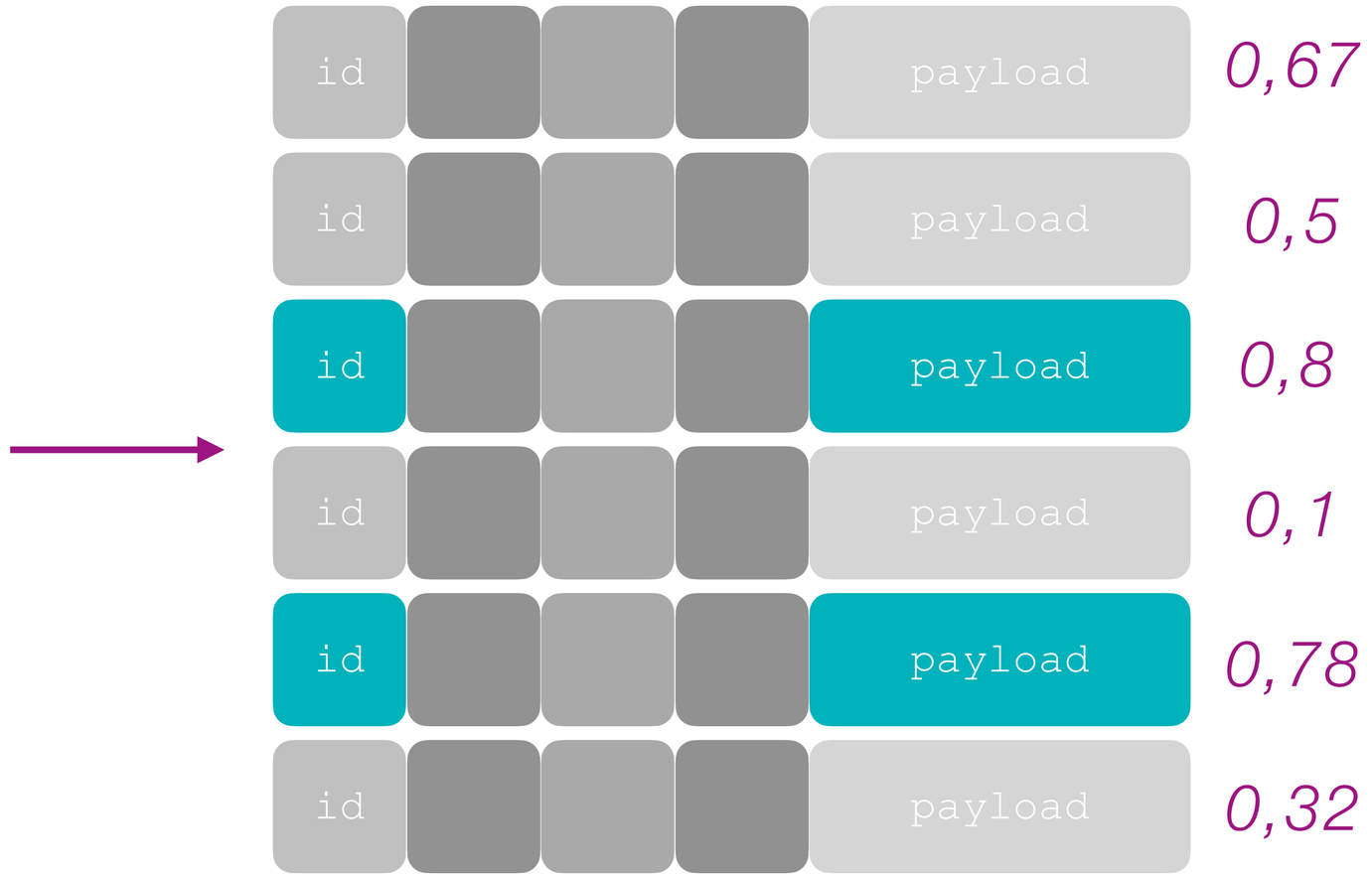
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the database

similarity search

query vector

[1, 0,6, 3 ...]

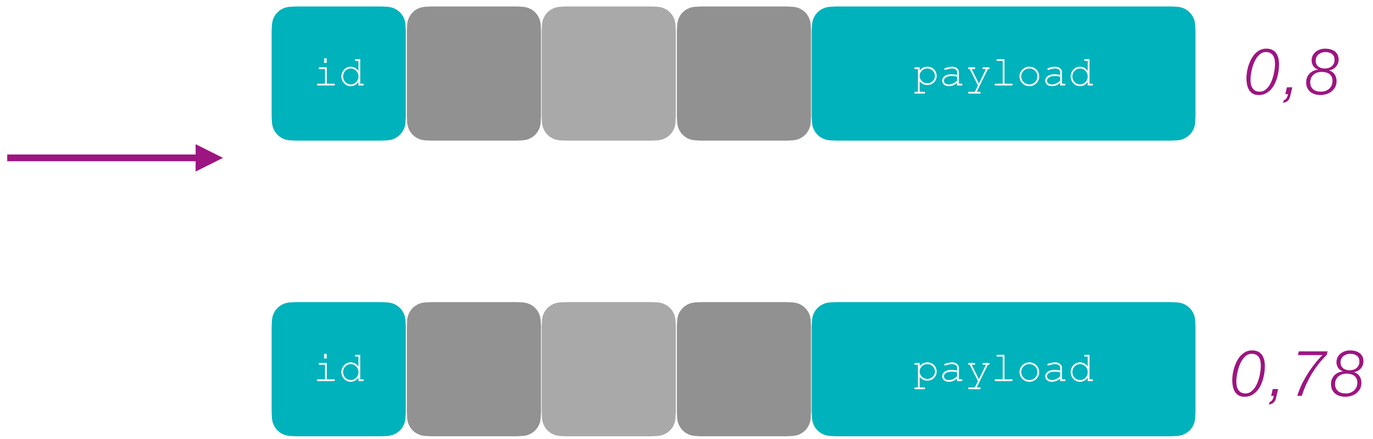


the database

similarity search

query vector

[1, 0,6, 3 ...]



the database

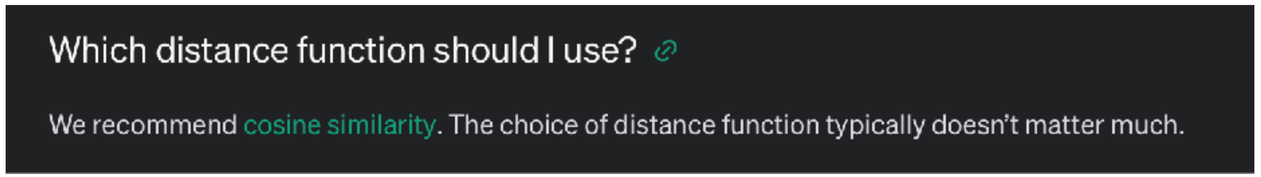
similarity search

embedding model for Data and Query must be the same

match the distance function you use to the one used to train the vector embedding model

source Oracle

Dot Product | **Cosine Similarity** | *Euclidean Distance*
Manhattan Distance | *Hammering Distance* | *Jaccard Distance*



or pick one and see what happens

source OpenAI

Vector DBs





Elasticsearch





redis

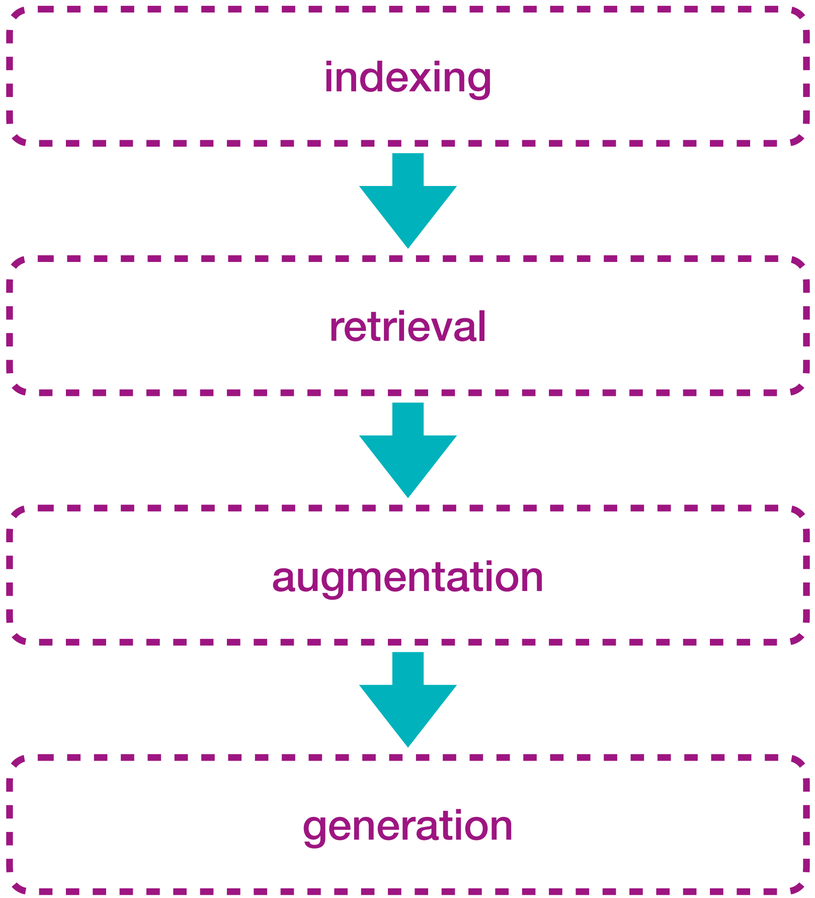


...



RAG

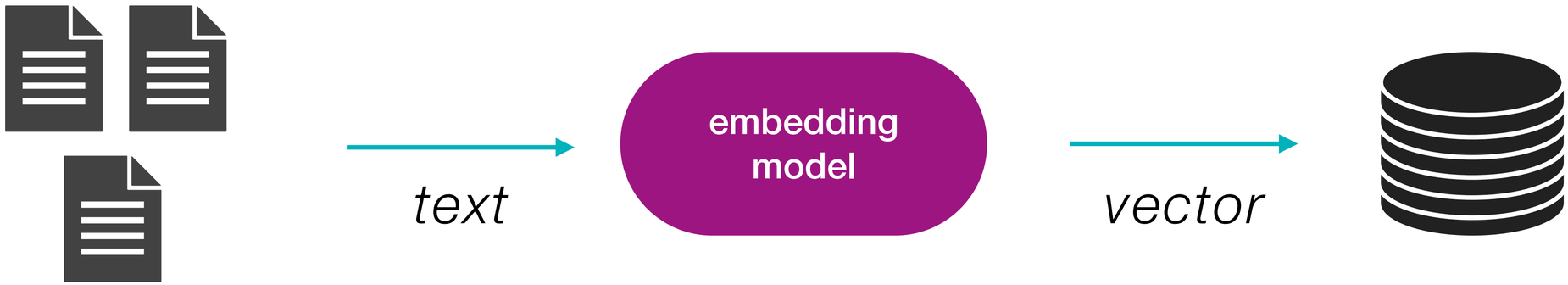
Retrieval augmented generation



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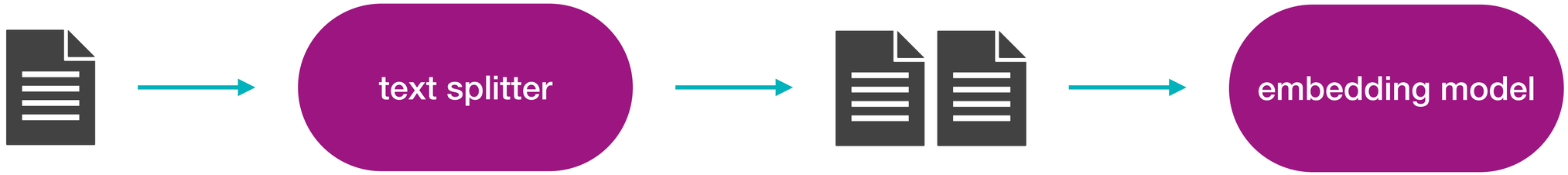
RAG

indexing



RAG

indexing

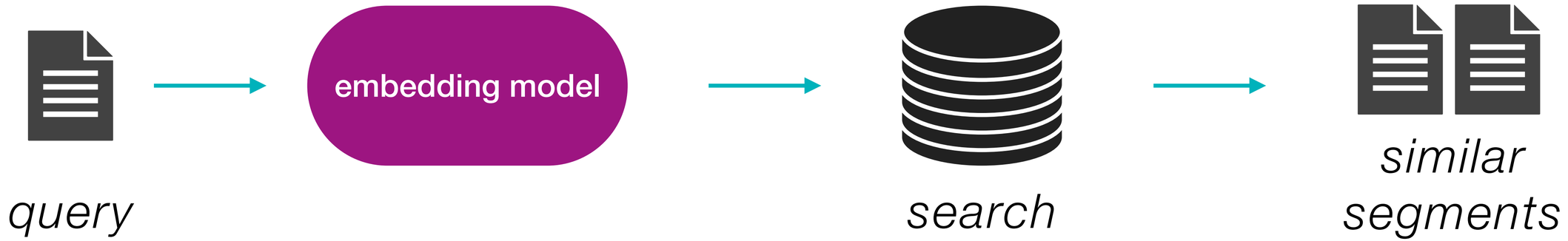


document

segments

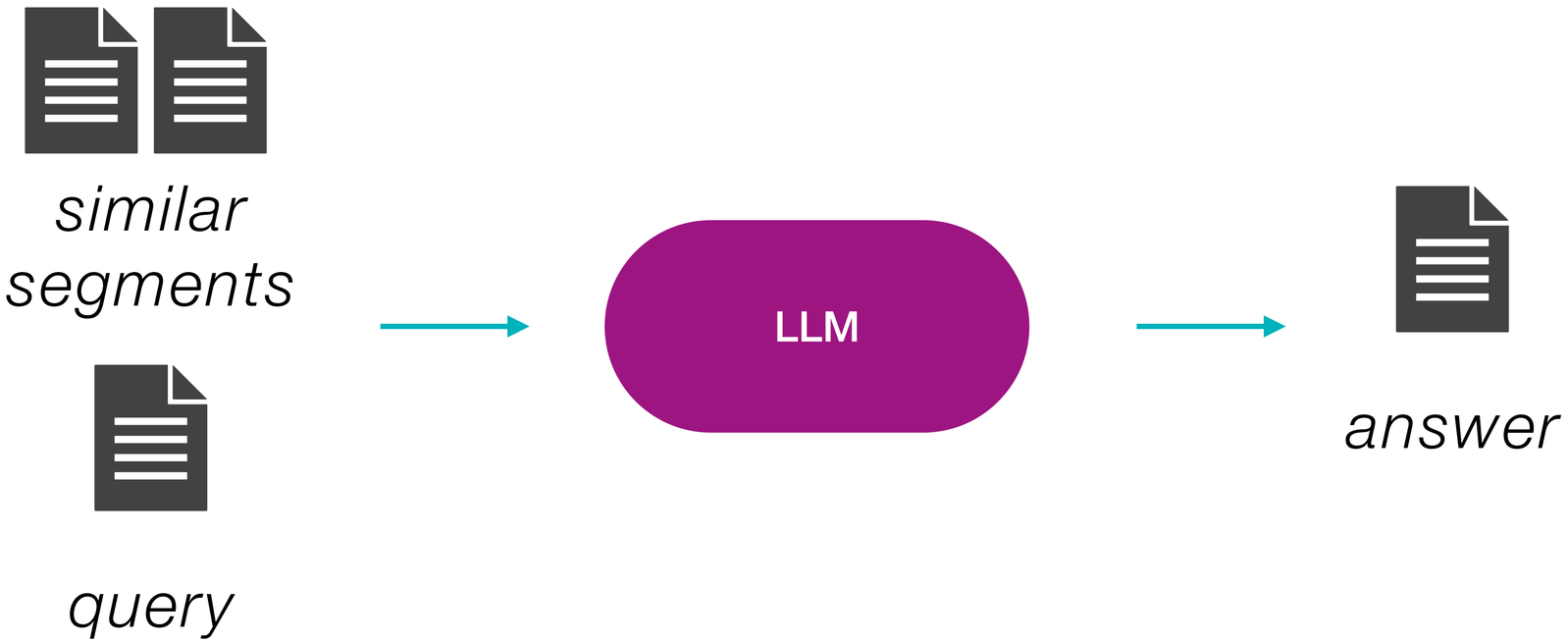
RAG

retrieval



RAG

augmentation & generation



demo



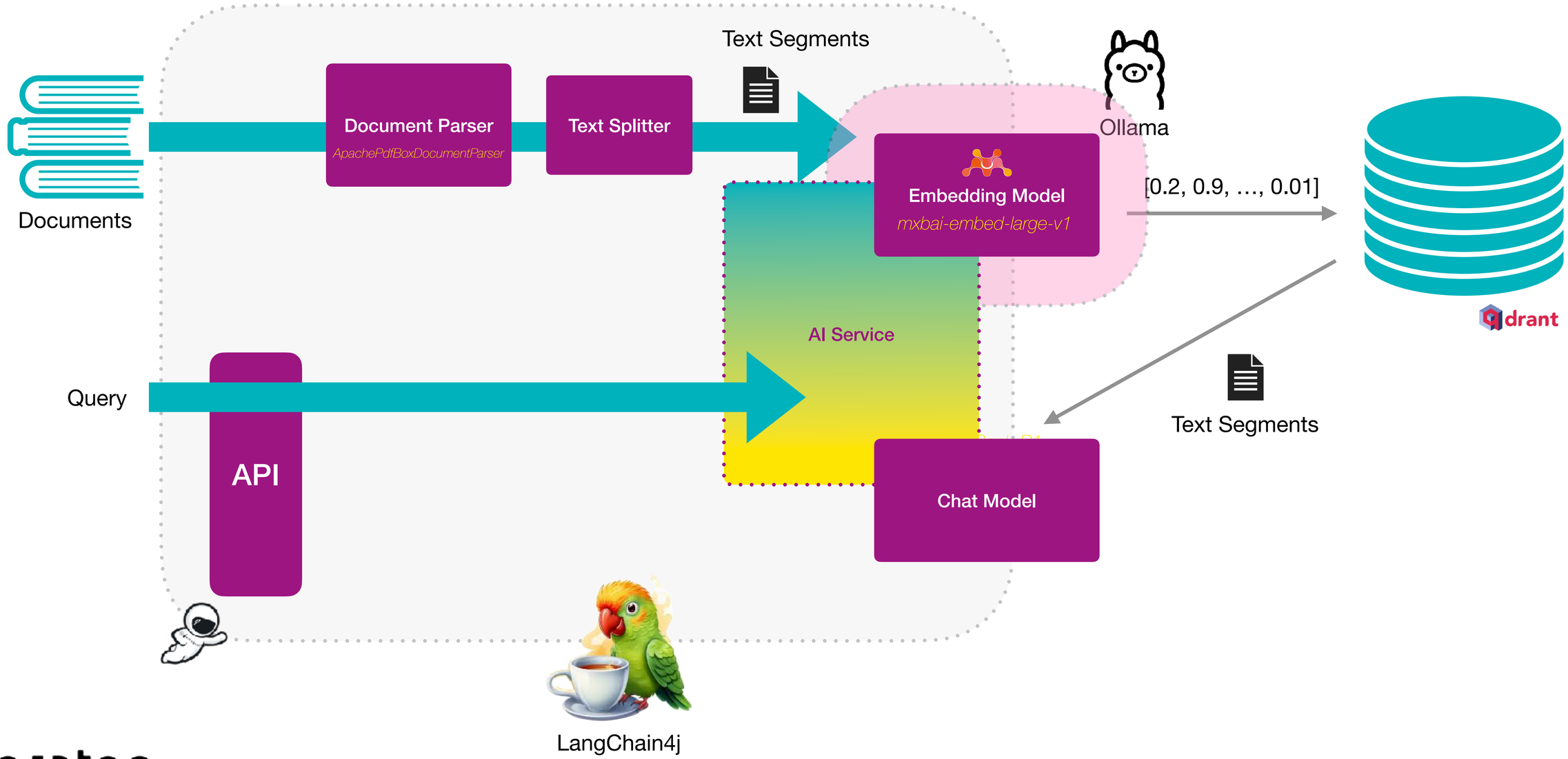
https://docs.langchain4j.dev/

Claude 4.5 Sonnet

mxbai-embed-large-v1

RAG

Demo architecture



Thank you

The Emperor Protects



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